

Practice Questions for 2024-10-28-Farkas

Generated by Scribe.AI

January 3, 2025

1 Questions

1. Consider Farkas' Lemma as presented in Slide 1. Suppose we have a system of inequalities $Ax \leq b$ where A is a 3×2 matrix and b is a 3×1 vector. If this system has no solution, what can we definitively conclude about the dimensions of a vector y satisfying the conditions $A^T y = 0$, $y \geq 0$, and $b^T y < 0$, and why?
 - a. y is a 2×1 vector.
 - b. y is a 3×1 vector.
 - c. y is a 2×3 vector.
 - d. y is a 3×2 vector.
 - e. The dimensions of y cannot be determined from the given information.
2.
 - a. Consider the system of inequalities $Ax \leq b$, where A is a $m \times n$ matrix and b is a $m \times 1$ vector. According to Farkas' Lemma (Slide 1), if this system has no solution, what can we conclude about the existence of a vector y satisfying $A^T y = 0$, $y \geq 0$, and $b^T y < 0$?
 - A. Such a vector y always exists.
 - B. Such a vector y may or may not exist.
 - C. Such a vector y never exists.
 - D. The existence of y depends on the specific values of A and b .
 - E. The existence of y depends on whether $m > n$ or $m < n$.
 - b. Suppose we have a system $Ax = b$ where A is a 5×3 matrix and b is a 5×1 vector. According to one version of Farkas' Lemma (Slide 2), if there is no non-negative solution x (i.e., $x \geq 0$), what can we conclude about the existence of a vector y such that $y^T A \geq 0^T$ and $y^T b < 0$?
 - A. Such a vector y always exists.
 - B. Such a vector y may or may not exist.
 - C. Such a vector y never exists.
 - D. The existence of y depends on the specific values of A and b .
 - E. The existence of y depends on the rank of matrix A .
 - c. Consider the three variants of Farkas' Lemma presented in Slide 4. Which variant directly addresses the solvability of the system $Ax \leq b$ without any non-negativity constraints on x ?
 - A. Variant (i)
 - B. Variant (ii)

- C. Variant (iii)
 - D. Variants (i) and (ii)
 - E. None of the variants directly address this.
3. Consider Farkas' Lemma (Slide 1). If the system $Ax \leq b$ has no solution, where A is a $m \times n$ matrix and b is a $m \times 1$ vector, what can we conclude about the vector y satisfying $A^T y = 0$, $y \geq 0$, and $b^T y < 0$? Specifically, what is the dimension of y ?
- a. The dimension of y is $n \times 1$.
 - b. The dimension of y is $n \times m$.
 - c. The dimension of y is $m \times 1$.
 - d. The dimension of y is $m \times n$.
 - e. The dimension of y cannot be determined from the given information.
4. Consider the system $Ax = b$, where A is a 5×3 matrix and b is a 5×1 vector (Slide 2). According to Farkas' Lemma, if there is no non-negative solution x ($x \geq 0$), what can we conclude about the vector y such that $y^T A \geq 0^T$ and $y^T b < 0$? What is the dimension of y ?
- a. The dimension of y is 3×1 .
 - b. The dimension of y is 3×5 .
 - c. The dimension of y is 5×1 .
 - d. The dimension of y is 5×3 .
 - e. The dimension of y cannot be determined.
5. Referring to Slide 4, which variant of Farkas' Lemma directly addresses the solvability of the system $Ax \leq b$ without any non-negativity constraints on x ?
- a. Variant (i)
 - b. Variant (ii)
 - c. Variant (iii)
 - d. None of the variants address this directly.
 - e. Variants (i) and (ii) both address this.
6. Suppose we have a system of inequalities $Ax \leq b$, where A is a 3×2 matrix and b is a 3×1 vector. If this system has no solution, what are the dimensions of the vector y that satisfies $A^T y = 0$, $y \geq 0$, and $b^T y < 0$ (Slide 1), and what is the interpretation of this result in the context of Farkas' Lemma?
- a. y is 2×1 ; it indicates the system is infeasible.
 - b. y is 3×1 ; it indicates the system is infeasible.
 - c. y is 2×1 ; it indicates the system is feasible.
 - d. y is 3×1 ; it indicates the system is feasible.
 - e. The dimensions of y cannot be determined.

7. a. Consider Farkas' Lemma (Slide 1): The system $Ax \leq b$ has no solutions if and only if there exists a vector y such that $A^T y = 0$, $y \geq 0$, and $b^T y < 0$. Suppose A is a $m \times n$ matrix and b is a $m \times 1$ vector. What is the dimension of the vector y , and why?
- A. $n \times 1$
 - B. $m \times 1$
 - C. $n \times m$
 - D. $m \times n$
 - E. The dimension of y is not determined by the dimensions of A and b .
- b. Consider the alternative version of Farkas' Lemma in Slide 2: Exactly one of the following holds: (F1) There exists $x \in R^n$ such that $Ax = b$ and $x \geq 0$; (F2) There exists $y \in R^m$ such that $y^T A \geq 0^T$ and $y^T b < 0$. If A is a 5×3 matrix and b is a 5×1 vector, what are the dimensions of x and y , and why?
- A. x is 3×1 , y is 3×1
 - B. x is 5×1 , y is 5×1
 - C. x is 3×1 , y is 5×1
 - D. x is 5×1 , y is 3×1
 - E. x is 3×5 , y is 5×3
- c. Referring to Slide 4, which variant of Farkas' Lemma directly addresses the solvability of the system $Ax \leq b$ without any non-negativity constraints on x ?
- A. Variant (i)
 - B. Variant (ii)
 - C. Variant (iii)
 - D. Variants (i) and (ii)
 - E. None of the variants directly address this.

2 Answers

1. Consider Farkas' Lemma as presented in Slide 1. Suppose we have a system of inequalities $Ax \leq b$ where A is a 3×2 matrix and b is a 3×1 vector. If this system has no solution, what can we definitively conclude about the dimensions of a vector y satisfying the conditions $A^T y = 0$, $y \geq 0$, and $b^T y < 0$, and why?
 - a. Incorrect. The dimensions of y are determined by the dimensions of A^T , which is the transpose of A .
 - b. The correct answer is B. Since A is a 3×2 matrix, its transpose A^T is a 2×3 matrix. The equation $A^T y = 0$ implies that y must have 3 rows to allow for matrix multiplication. The condition $y \geq 0$ indicates that y is a column vector, thus y is a 3×1 vector.
 - c. Incorrect. y must be a column vector to satisfy the condition $A^T y = 0$.
 - d. Incorrect. The dimensions of y are determined by the dimensions of A^T , which is the transpose of A .
 - e. Incorrect. The dimensions of y are directly determined by the dimensions of A and the requirement that $A^T y = 0$ be a valid matrix multiplication.
2.
 - a. Consider the system of inequalities $Ax \leq b$, where A is a $m \times n$ matrix and b is a $m \times 1$ vector. According to Farkas' Lemma (Slide 1), if this system has no solution, what can we conclude about the existence of a vector y satisfying $A^T y = 0$, $y \geq 0$, and $b^T y < 0$?
 - A. Farkas' Lemma states that if the system $Ax \leq b$ has no solution, then there exists a vector y such that $A^T y = 0$, $y \geq 0$, and $b^T y < 0$. This is a direct consequence of the lemma's statement.
 - B. This is incorrect because Farkas' Lemma provides a definitive statement about the existence of y when $Ax \leq b$ has no solution.
 - C. This is incorrect; Farkas' Lemma guarantees the existence of such a vector y when the original system has no solution.
 - D. This is incorrect. The lemma's statement is independent of the specific values of A and b , provided the system $Ax \leq b$ is infeasible.
 - E. This is incorrect. The dimensions of A and b influence the dimensions of y , but the lemma's core statement holds regardless of the relationship between m and n .
 - b. Suppose we have a system $Ax = b$ where A is a 5×3 matrix and b is a 5×1 vector. According to one version of Farkas' Lemma (Slide 2), if there is no non-negative solution x (i.e., $x \geq 0$), what can we conclude about the existence of a vector y such that $y^T A \geq 0^T$ and $y^T b < 0$?
 - A. According to the version of Farkas' Lemma on Slide 2, if the system $Ax = b$ has no non-negative solution, then there exists a vector y such that $y^T A \geq 0^T$ and $y^T b < 0$. This is a direct application of the lemma.
 - B. This is incorrect; Farkas' Lemma provides a definitive statement about the existence of y when $Ax = b$ has no non-negative solution.
 - C. This is incorrect; the lemma guarantees the existence of such a vector y when the original system has no non-negative solution.
 - D. This is incorrect. The lemma's statement is independent of the specific values of A and b , provided the system $Ax = b$ has no non-negative solution.
 - E. While the rank of A is relevant to the solvability of $Ax = b$, the lemma's statement about the existence of y is independent of the rank of A .

- c. Consider the three variants of Farkas' Lemma presented in Slide 4. Which variant directly addresses the solvability of the system $Ax \leq b$ without any non-negativity constraints on x ?
 - A. Variant (i) deals with $Ax = b$ and $x \geq 0$.
 - B. Variant (ii) deals with $Ax \leq b$ and $x \geq 0$.
 - C. Variant (iii) explicitly deals with the existence of a solution to $Ax \leq b$ without any restriction on the sign of x .
 - D. Both (i) and (ii) impose non-negativity constraints on x .
 - E. This is incorrect; variant (iii) directly addresses this case.

3. Consider Farkas' Lemma (Slide 1). If the system $Ax \leq b$ has no solution, where A is a $m \times n$ matrix and b is a $m \times 1$ vector, what can we conclude about the vector y satisfying $A^T y = 0$, $y \geq 0$, and $b^T y < 0$? Specifically, what is the dimension of y ?
 - a. Incorrect. The dimension of y is determined by the dimensions of A^T , which is $n \times m$. The equation $A^T y = 0$ requires y to have m entries.
 - b. Incorrect. This is not a valid matrix dimension for the equation $A^T y = 0$ to hold.
 - c. The correct answer is C. Farkas' Lemma states that if $Ax \leq b$ has no solution, then there exists a vector y such that $A^T y = 0$, $y \geq 0$, and $b^T y < 0$. Since A is an $m \times n$ matrix, A^T is an $n \times m$ matrix. For the equation $A^T y = 0$ to be consistent, y must have dimension $m \times 1$.
 - d. Incorrect. This is not a valid matrix dimension for the equation $A^T y = 0$ to hold.
 - e. Incorrect. The dimension of y is directly determined by the dimensions of A and the requirement that $A^T y = 0$ be a valid matrix equation.

4. Consider the system $Ax = b$, where A is a 5×3 matrix and b is a 5×1 vector (Slide 2). According to Farkas' Lemma, if there is no non-negative solution x ($x \geq 0$), what can we conclude about the vector y such that $y^T A \geq 0^T$ and $y^T b < 0$? What is the dimension of y ?
 - a. Incorrect. y must have 5 entries to make $y^T A$ a valid matrix multiplication.
 - b. Incorrect. This is not a valid matrix dimension for the product $y^T A$ to be defined.
 - c. The correct answer is C. Farkas' Lemma (Slide 2) states that if $Ax = b$ has no non-negative solution, then there exists a vector y such that $y^T A \geq 0^T$ and $y^T b < 0$. Since A is a 5×3 matrix, y^T must be a 1×5 row vector for the product $y^T A$ to be defined. Therefore, y is a 5×1 column vector.
 - d. Incorrect. This is not a valid matrix dimension for the product $y^T A$ to be defined.
 - e. Incorrect. The dimension of y is determined by the requirement that $y^T A$ be a valid matrix product.

5. Referring to Slide 4, which variant of Farkas' Lemma directly addresses the solvability of the system $Ax \leq b$ without any non-negativity constraints on x ?
 - a. Incorrect. Variant (i) deals with $Ax = b$ and non-negative x .
 - b. Incorrect. Variant (ii) deals with $Ax \leq b$ and non-negative x .
 - c. The correct answer is C. Variant (iii) of Farkas' Lemma (Slide 4) states that the system $Ax \leq b$ has a solution (without any non-negativity constraint on x) if and only if every non-negative $y \in R^m$ with $y^T A = 0^T$ also satisfies $y^T b \geq 0$.

- d. **Incorrect.** Variant (iii) directly addresses this case.
- e. **Incorrect.** Only variant (iii) addresses the solvability of $Ax \leq b$ without non-negativity constraints on x .
6. Suppose we have a system of inequalities $Ax \leq b$, where A is a 3×2 matrix and b is a 3×1 vector. If this system has no solution, what are the dimensions of the vector y that satisfies $A^T y = 0$, $y \geq 0$, and $b^T y < 0$ (Slide 1), and what is the interpretation of this result in the context of Farkas' Lemma?
- a. **Incorrect.** The dimension of y must match the number of rows in A , which is 3.
- b. The correct answer is B. Since A is a 3×2 matrix, A^T is a 2×3 matrix. For the equation $A^T y = 0$ to be consistent, y must be a 3×1 vector. Farkas' Lemma states that if $Ax \leq b$ has no solution, then such a y exists, confirming the infeasibility of the original system.
- c. **Incorrect.** The existence of such a y indicates infeasibility, not feasibility.
- d. **Incorrect.** While the dimension is correct, the interpretation is wrong; the existence of y signifies infeasibility.
- e. **Incorrect.** The dimensions of y are directly determined by the dimensions of A .
7. a. Consider Farkas' Lemma (Slide 1): The system $Ax \leq b$ has no solutions if and only if there exists a vector y such that $A^T y = 0$, $y \geq 0$, and $b^T y < 0$. Suppose A is a $m \times n$ matrix and b is a $m \times 1$ vector. What is the dimension of the vector y , and why?
- A. **Incorrect.** The transpose of A is $n \times m$, so y cannot be $n \times 1$ for the multiplication $A^T y$ to be defined.
- B. The dimension of y is $m \times 1$. The equation $A^T y = 0$ involves the transpose of A , which is an $n \times m$ matrix. Since y is multiplied by A^T on the left, y must be a column vector with m rows to allow for matrix multiplication. Therefore, the dimension of y is $m \times 1$.
- C. **Incorrect.** A^T is $n \times m$, so y cannot be $n \times m$ for the multiplication $A^T y$ to be defined.
- D. **Incorrect.** A^T is $n \times m$, so y cannot be $m \times n$ for the multiplication $A^T y$ to be defined.
- E. **Incorrect.** The dimension of y is directly determined by the dimensions of A and the requirement that $A^T y$ is a well-defined matrix multiplication.
- b. Consider the alternative version of Farkas' Lemma in Slide 2: Exactly one of the following holds: (F1) There exists $x \in R^n$ such that $Ax = b$ and $x \geq 0$; (F2) There exists $y \in R^m$ such that $y^T A \geq 0^T$ and $y^T b < 0$. If A is a 5×3 matrix and b is a 5×1 vector, what are the dimensions of x and y , and why?
- A. **Incorrect.** The dimensions of x and y are not both 3×1 .
- B. **Incorrect.** The dimension of x is not 5×1 .
- C. In (F1), $Ax = b$ requires A (5×3) multiplied by x to result in a 5×1 vector b . Thus, x must be 3×1 . In (F2), $y^T A$ must be a 1×3 row vector, and $y^T b$ must be a scalar. Since A has 5 rows, y must be a 5×1 column vector.
- D. **Incorrect.** The dimension of y is not 3×1 .
- E. **Incorrect.** The dimensions of x and y are not 3×5 and 5×3 respectively.
- c. Referring to Slide 4, which variant of Farkas' Lemma directly addresses the solvability of the system $Ax \leq b$ without any non-negativity constraints on x ?
- A. **Incorrect.** Variant (i) deals with $Ax = b$ and $x \geq 0$.

- B. **Incorrect.** Variant (ii) deals with $Ax \leq b$ and $x \geq 0$.
- C. Variant (iii) states: "The system $Ax \leq b$ has a solution if and only if every nonnegative $y \in R^m$ with $y^T A = 0^T$ also satisfies $y^T b \geq 0$." This is the only variant that doesn't impose a non-negativity constraint on x .
- D. **Incorrect.** Only variant (iii) addresses the solvability of $Ax \leq b$ without non-negativity constraints on x .
- E. **Incorrect.** Variant (iii) directly addresses this.